

Unrolled Iterative Reconstruction with Calibrated Uncertainty for Low-Dose CT: A Reference Method for the PWM Benchmark

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Abstract

We present an open-source reference reconstruction method for low-dose CT designed as a **credible, improvable, and permissively-licensed** community baseline rather than a single-dataset state-of-the-art point. The method is an unrolled iterative reconstruction with 10 data-consistency steps interleaved with a weight-shared U-Net denoiser; uncertainty is quantified via a five-model deep ensemble; the forward model is the open `ct_radon` implementation maintained as part of the PWM modality engine, so any user replicating the architecture inherits a Radon transform validated against PWM’s 172-modality test suite. We evaluate on the multi-vendor PWM-LDCT benchmark across four dose levels (10%, 25%, 50%, 100%) and report five complementary axes that are rarely co-located in published methods: reconstruction quality (PSNR / SSIM / LPIPS), downstream task performance (lung-nodule detection AUC), cross-vendor generalization (leave-one-vendor-out), per-pixel uncertainty calibration (Spearman ρ of predicted variance vs observed error), and signal-equivalence credentials per the framework of [to be confirmed \(2027b\)](#). The method ships under Apache 2.0 with a content-addressed Docker RunBundle on IPFS and a verified L4 cert on the PWM blockchain registry. Its architecture is *decomposably improvable*: each of the four components (denoiser, forward model, iteration schedule, UQ method) can be swapped independently, inviting follow-on contributions to compete on individual axes rather than re-derive the full stack.

Keywords: Low-dose CT Unrolled iterative reconstruction Uncertainty quantification Cross-vendor generalization Reproducible benchmark.

1 Introduction

Despite years of method papers reporting fractional-dB PSNR gains over previous best ([Chen et al., 2017](#); [Wang et al., 2020](#); [to be confirmed at submission, 2024](#)), the low-dose CT field has no single reference baseline. The obstacle is structural: published methods optimize for performance on the dataset used during development (typically AAPM-2016 ([AAPM, 2016](#)) on Siemens hardware) and rarely report **(i)** cross-vendor generalization, **(ii)** per-pixel uncertainty calibration, **(iii)** downstream task performance, or **(iv)** open-source code with permissive licensing. A deployer picking a method for a different vendor, or one needing per-case diagnostic confidence, has no clear choice.

We present an open-source reference method explicitly designed for these four properties, positioned as the seed entry of a permanent benchmark leaderboard ([to be confirmed, 2028](#)).

Position vs vendor IR. Vendor-deployed iterative reconstruction (Siemens *Admirer*, GE *Veo*, Canon *AIDR3D*) is FDA-cleared but closed: weights, training data, and inference code are vendor-proprietary; cross-vendor evaluation is therefore not possible. Open methods such as RED-CNN ([Chen et al., 2017](#)) are reproducible but lack UQ and cross-vendor scope. We aim to combine the architectural class of vendor IR with the reproducibility of academic methods.

Architecture rationale. We use unrolled iterative reconstruction (Adler and Öktem, 2018; Mardani et al., 2018) for three reasons. **(1) Physical prior:** embedding the (exact) Radon operator in the iteration loop guarantees data consistency that end-to-end networks must learn from data. **(2) Decomposable improvability:** a follow-up can independently swap the denoiser, forward model, iteration schedule, or UQ method — each yielding a publishable contribution that improves on this baseline without invalidating it. Pure transformer or pure diffusion architectures do not offer this decomposability (to be confirmed at submission, 2024). **(3) Regulatory transferability:** iterative methods with explicit data-consistency loops have FDA-cleared precedent in vendor IR; an architecturally analogous open method is more transferable to regulated deployment than a pure neural approach.

Why “credible and improvable” beats “state-of-the-art”. A reference method that wins by 0.3 dB on a single vendor is too fragile to anchor a community benchmark: the next year’s method on a slightly different dataset displaces it, and the cumulative leaderboard has no stable origin. We instead target a competitive, mid-range performer with four desirable properties present, designed for follow-on improvement on independent axes. This is the same logic that made BERT-base a productive baseline relative to a hypothetical singularly-tuned competitor; the open question of which axis to improve next is itself a contribution to the field.

Contributions.

1. An open-source unrolled iterative architecture with weight-shared U-Net denoiser, TV warm start, and five-model deep ensemble UQ, sized to fit in < 24 GB GPU memory (Section 3).
2. Cross-vendor generalization evaluation: leave-one-vendor-out on the PWM-LDCT benchmark (to be confirmed, 2027a) (Section 4).
3. Per-pixel uncertainty calibration: Spearman correlation between predicted variance and observed reconstruction error (Section 5).
4. Five-tuple signal-equivalence credentials per to be confirmed (2027b) at 25% and 10% dose (Section 5.1).
5. Release under Apache 2.0 with a reproducible Docker RunBundle on IPFS and a corresponding L4 cert on PWM mainnet.

2 Related work

Deep learning for low-dose CT. The 2017 introduction of RED-CNN (Chen et al., 2017) established encoder-decoder CNNs as a competitive low-dose denoising baseline. WGAN-VGG (Yang et al., 2018) and subsequent perceptual-loss methods improved visual quality at similar PSNR. CTformer (Wang et al., 2023) brought transformer self-attention to the task and reported ~ 0.5 dB PSNR gains. Diffusion- and score-based methods (to be confirmed at submission, 2024) achieve state-of-the-art PSNR on AAPM-2016 at the cost of substantial inference latency. Wang et al. (2020) survey the field. Common to all is single-vendor evaluation on AAPM-2016 or LIDC-IDRI; cross-vendor generalization is not reported.

Unrolled iterative reconstruction. Learned primal-dual schemes (Adler and Öktem, 2018) and learned proximal gradient methods (Mardani et al., 2018) embed a learned denoiser inside a classical iterative loop. Weight sharing across iterations reduces parameter count and is a known implicit regularizer (Adler and Öktem, 2018). The architectural choice is interpretable, decomposable, and architecturally analogous to FDA-cleared vendor iterative reconstruction — properties we explicitly exploit (Section 3.1).

Uncertainty and cross-domain generalization. Monte-Carlo dropout (Gal and Ghahramani, 2016) is cheap but poorly calibrated under distribution shift (Ovadia et al., 2019); deep ensembles (Lakshminarayanan et al., 2017) consistently outperform dropout on calibration and have been extended to medical segmentation (Mehrtash et al., 2020) and CT reconstruction (Barbano et al., 2022). For distribution shift, domain-adversarial training (Ganin et al., 2016), BN adaptation (Li et al., 2017), and test-time training (Sun

et al., 2020) target the general problem; cross-scanner medical-image segmentation has been studied (Karani et al., 2018), but cross-vendor evaluation for CT reconstruction is largely absent from the literature — the gap this paper fills.

Benchmarks and reproducibility. AAPM-2016 (AAPM, 2016) and Mayo LDCT-PD (McCollough et al., 2017) are the field’s two real-paired-low-dose CT datasets; both are Siemens-only. The PWM-LDCT benchmark (to be confirmed, 2027a) extends to multiple vendors. The signal-equivalence framework (to be confirmed, 2027b) provides the credential format we report against.

3 Method

3.1 Architecture overview

The pipeline is summarized in Figure 1.

[TODO: FIGURE: Architecture diagram. The rendered figure should show: (left) input low-dose sinogram y ; (center) the unrolled iteration loop with $K = 10$ blocks, each block alternating a data-consistency step ($x^{(k+1/2)} = x^{(k)} - \tau \mathcal{R}^\dagger(\mathcal{R}x^{(k)} - y)$) and a learned weight-shared U-Net denoiser f_θ ; (right) the deep-ensemble head producing per-pixel mean $\hat{x}(y)$ and per-pixel standard deviation $\sigma(y)$ from $M = 5$ independently-trained instances. A side annotation should highlight that the forward operator $\mathcal{R}$ is the upstream PWM `ct_radon` module, that τ is a learned scalar shared across iterations, and that the denoiser weights θ are shared across all K iterations within one ensemble member.]

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3.2 Forward model

Let $\mathcal{R} : \mathbb{R}^{H \times W} \rightarrow \mathbb{R}^{n_{\text{views}} \times n_{\text{dets}}}$ denote the Radon forward operator and $\mathcal{R}^\dagger$ its filtered back-projection adjoint. Both are implemented in `pwm_core.contrib.modalities.ct_radon` (PWM Protocol, 2026), the validated forward-model implementation used across PWM’s 172-modality engine. We do not re-implement; using the upstream implementation guarantees that bug-fixes propagate uniformly and that any user of our reference method inherits a Radon transform validated against the full PWM modality test suite.

3.3 Unrolled iteration

Given a low-dose sinogram $y \in \mathbb{R}^{n_{\text{views}} \times n_{\text{dets}}}$, we estimate the reconstruction $\hat{x} \in \mathbb{R}^{H \times W}$ via K iterations of a data-consistency step interleaved with a learned denoiser f_θ :

We use $K = 10$ iterations; ablation results for $K \in \{5, 10, 15, 20\}$ are in Section 5 Table 1. The denoiser f_θ is a U-Net with ≈ 4 M parameters whose weights are shared across iterations; this both reduces memory and acts as an implicit regularizer (the same f_θ must denoise at all iteration depths, preventing iteration-specific overfitting).

The step size τ is a learned scalar, initialized at $0.5/\|\mathcal{R}\|^2$ (the conventional Landweber-stable value). End-to-end optimization with mean absolute error loss on Hounsfield-unit values:

Algorithm 1 Unrolled iterative reconstruction

```
1:  $x^{(0)} \leftarrow \text{TV-FBP}(y)$  ▷ TV-regularized warm start
2: for  $k = 0, 1, \dots, K-1$  do
3:    $x^{(k+1/2)} \leftarrow x^{(k)} - \tau \cdot \mathcal{R}^\dagger(\mathcal{R}(x^{(k)}) - y)$  ▷ Data consistency
4:    $x^{(k+1)} \leftarrow f_\theta(x^{(k+1/2)})$  ▷ Learned denoising
5: end for
6: return  $\hat{x} \leftarrow x^{(K)}$ 
```

[TODO: TABLE: Iteration-count ablation: PSNR / SSIM on PWM-LDCT test split for $K \in \{5, 10, 15, 20\}$]

Table 1: Iteration-count ablation: PSNR / SSIM on PWM-LDCT test split for $K \in \{5, 10, 15, 20\}$ (placeholder; populate when data lands).

$$\mathcal{L}(\theta, \tau) = \frac{1}{N} \sum_{i=1}^N \|\hat{x}_i(\theta, \tau; y_i) - x_i^{\text{ref}}\|_1 \quad (1)$$

where x_i^{ref} is the patient’s full-dose reconstruction (the reference image).

3.4 Uncertainty quantification

We use a five-model deep ensemble (Lakshminarayanan et al., 2017): five independently-initialized and independently-trained instances of the same architecture, differing only in random seed for weight initialization and data ordering. At inference time, given a low-dose sinogram y , each ensemble member produces a reconstruction $\hat{x}_m(y)$ for $m = 1, \dots, 5$. The point estimate is the mean and the per-pixel uncertainty is the standard deviation:

$$\hat{x}(y) = \frac{1}{5} \sum_{m=1}^5 \hat{x}_m(y), \quad \sigma(y)_{ij} = \sqrt{\frac{1}{5} \sum_{m=1}^5 (\hat{x}_m(y)_{ij} - \hat{x}(y)_{ij})^2}. \quad (2)$$

We choose deep ensembles over Monte-Carlo dropout because (i) ensembles consistently outperform dropout on calibration in regression benchmarks (Ovadia et al., 2019), (ii) inference cost is bounded ($5 \times$ a single model rather than T -stochastic), and (iii) the inference cost can be amortized across many predictions on the same GPU.

What the uncertainty is for. A clinically actionable UQ map distinguishes patient cases for which the reconstruction is reliable (low $\sigma(y)$ everywhere in the diagnostic region) from cases for which the reconstruction is uncertain enough to warrant either a higher-dose retake, a second-reader review, or a deferred diagnosis. A well-calibrated reference method therefore reports not only per-pixel variance but its *agreement with per-pixel error* on a held-out test set (Section 5). A high Spearman correlation gives radiologists evidence that the variance map is informative; a low correlation flags that the ensemble disagreement does not track the actual reconstruction quality and the UQ should not be used for triage. The PWM-LDCT benchmark’s paired full-dose ground truth makes this evaluation possible at scale.

3.5 Training data and augmentation

Training data: full PWM-LDCT (to be confirmed, 2027a) training split with paired (full-dose, low-dose) reconstructions at $r \in \{0.10, 0.25, 0.50\}$. We mix all three dose levels in each minibatch with weights (0.4, 0.4, 0.2), biased toward the most clinically relevant levels.

Augmentation: random flip ($p = 0.5$), random rotation in $[-5^\circ, 5^\circ]$, random window-level shift in $[-50, 50]$ HU. No augmentation that violates physical consistency (no random noise injection on top of the already-noisy low-dose, no patch shuffling).

3.6 Implementation details

The U-Net denoiser has four encoder / four decoder stages with (32, 64, 128, 256) channels (GELU, GroupNorm, bilinear upsampling), ≈ 4 M parameters per ensemble member. The data-consistency step uses a single learned scalar τ initialized at $0.5/\|\mathcal{R}\|_2^2$. Optimizer is AdamW (Loshchilov and Hutter, 2019), cosine LR schedule with 5% warmup; training uses batch size 4 (512×512 full slices) on a single A100. Ensemble seeds {42, 43, 44, 45, 46}; CuDNN deterministic; all RNGs seed-propagated. Full hyperparameter table, training compute disclosure, and ablation hypothesis statements are in Supplementary Tables S1–S2.

4 Experiments

Datasets. Training: PWM-LDCT (to be confirmed, 2027a) training split ([TODO: N] patients, [TODO: N] scans). Evaluation: PWM-LDCT test split ([TODO: N] patients), AAPM-2016 (AAPM, 2016) held-out scans ([TODO: N] patients), and LIDC-IDRI (Armato III et al., 2011) sub-sample ([TODO: N] patients) for downstream task validation.

Baselines. Three baseline reconstruction methods, reproduced within ≤ 0.5 dB PSNR of their published numbers:

- RED-CNN (Chen et al., 2017): foundational encoder-decoder CNN, 2017.
- A transformer-based reconstruction ([TODO: select: CTformer / TransCT / Restormer-LDCT]).
- A diffusion / score-based reconstruction ([TODO: select: 2024 score-LDCT]).

Cross-vendor splits. We construct three leave-one-vendor-out splits: train-on-{GE, Canon}/test-on-Siemens; train-on-{Siemens, Canon}/test-on-GE; train-on-{Siemens, GE}/test-on-Canon. Each cross-vendor experiment uses the full training set with the held-out vendor’s patients removed (patient-level, not slice-level).

Metrics. (i) PSNR, SSIM, LPIPS (Zhang et al., 2018) on the test reconstructions; (ii) lung-nodule detection AUC at fixed FPR = 0.1, using a pre-trained nodule detector frozen across reconstruction methods; (iii) per-pixel uncertainty calibration — Spearman correlation of $\sigma(y)_{ij}$ against $|\hat{x}(y)_{ij} - x_i^{\text{ref}}|_{ij}$ on held-out test; (iv) inference latency (seconds per slice on a single NVIDIA A100), peak GPU memory; (v) 5-tuple signal-equivalence credentials per the framework of to be confirmed (2027b) at $r \in \{0.10, 0.25\}$ for lung-nodule detection, $\varepsilon = 0.02$, $\alpha = 0.05$, subpopulation $\Pi = \text{adult-chest}$ test split.

Ablations. Five ablations isolate each design choice: iteration count $K \in \{5, 10, 15, 20\}$; ensemble size $M \in \{1, 3, 5, 10\}$; weight-shared vs per-iteration denoiser; TV-FBP vs FBP vs learned warm-start; mixed-dose training vs $r = 0.25$ -only training. Hypotheses and detailed results in Supplementary Table S2.

Statistical reporting. All metrics reported with 95% bootstrap CIs ($B = 1000$) over test-set patients; cross-vendor PSNR drops compared with paired bootstrap tests at $\alpha = 0.05$; Holm-Bonferroni correction for multi-split tests.

5 Results

Reconstruction quality. Table 2 [TODO: populate] reports PSNR, SSIM, and LPIPS for each method at each dose level on the PWM-LDCT test split. The reference method achieves PSNR within [TODO: X] dB of the diffusion-based baseline at $r = 0.25$, while consuming [TODO: Y] $\times$ less inference time and using [TODO: Z] $\times$ less GPU memory.

Cross-vendor generalization. Table 3 reports the PSNR drop Δ when each method is trained without one vendor and evaluated on that held-out vendor’s test scans. The reference method’s Δ should be ≤ 2 dB; baselines are not designed for cross-vendor generalization and we expect $\Delta \approx 3\text{--}5$ dB.

Method	$r = 0.10$		$r = 0.25$		$r = 0.50$	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
FBP	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
RED-CNN	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Transformer	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Diffusion	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
PWM ref	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Table 2: Reconstruction quality on the PWM-LDCT test split. Values to fill at D9+270 after WS-3 Phase 3 (v1 deep ensemble) lands.

[TODO: TABLE: Cross-vendor PSNR drop (Δ , dB) under leave-one-vendor-out training; lower is better]

Table 3: Cross-vendor PSNR drop (Δ , dB) under leave-one-vendor-out training; lower is better (placeholder; populate when data lands).

Uncertainty calibration. The deep ensemble’s per-pixel standard deviation correlates with per-pixel error at Spearman $\rho =$ [TODO: populate]. A reliability diagram is provided as Figure 2.

[TODO: FIGURE: Reliability diagram: predicted per-pixel standard deviation (ensemble) vs observed per-pixel reconstruction error on held-out test]

Figure 2: Reliability diagram: predicted per-pixel standard deviation (ensemble) vs observed per-pixel reconstruction error on held-out test (placeholder; populate when data lands).

Downstream task performance. Lung-nodule detection AUC against a frozen detector is reported in Table 4. The downstream-task results may diverge from pixel-wise results: a method with higher PSNR may produce lower task AUC if it over-smooths small lesions (Kim et al., 2019). Showing both is the point.

5.1 Signal-equivalence credentials

Per the framework of to be confirmed (2027b), we report 5-tuple credentials for the reference method:

These credentials are the canonical artifact that future methods will report against. Reporting only PSNR/SSIM/LPIPS is insufficient; reporting only verdicts loses the CI; reporting both is the standard we propose.

6 Discussion

What this method is and is not. The PWM reference reconstruction is not state-of-the-art on the single-vendor AAPM-2016 benchmark, and we do not claim it is. It is the first open-source reconstruction method that ships with calibrated per-pixel UQ, validated cross-vendor generalization, and an Apache-2.0-licensed RunBundle anchored to a content-addressed benchmark hash. Each of these four properties has been studied in isolation (Lakshminarayanan et al., 2017; Karani et al., 2018; Chen et al., 2017); their combination in a single deployable artifact has not.

When to use it and how. The method is appropriate when the deployment target is a different vendor or scanner than the training data, when per-case diagnostic confidence is required for triage, when regulatory review requires open-source weights, or when subsequent improvements should be evaluated against a stable reference. It is *not* appropriate when single-vendor single-task PSNR is the only success criterion. The per-pixel UQ is intended for case-level triage (high $\bar{\sigma}$ over the diagnostic region $\Rightarrow$ flag for retake or second-reader

[TODO: TABLE: Lung-nodule detection AUC at fixed FPR = 0.1, per method, per dose level, using a frozen pre-trained detector]

Table 4: Lung-nodule detection AUC at fixed FPR = 0.1, per method, per dose level, using a frozen pre-trained detector (placeholder; populate when data lands).

r	Task	ε	α	Subpopulation	Verdict	CI
0.25	Lung nodule AUC	0.02	0.05	adult-chest test split	[TODO:]	[TODO:]
0.10	Lung nodule AUC	0.05	0.05	adult-chest test split	[TODO:]	[TODO:]
0.25	Liver lesion Dice	0.03	0.05	adult-abdomen test split	[TODO:]	[TODO:]

Table 5: Signal-equivalence credentials computed per [to be confirmed](#) (2027b) against full-dose FBP as the reference. Verdicts: **PASS** / **FAIL** / **INDETERMINATE**.

review), not for pixel-level CIs on individual structures; we recommend pairing it with the signal-equivalence credential ([to be confirmed](#), 2027b) so the variance flags individual cases while the credential certifies the cohort.

Limitations. (1) Vendor coverage in the released training set is **[TODO: N]** of the four major CT manufacturers; cross-vendor results to vendors not in training are extrapolation, not measurement. (2) Training-dose mix $r \in \{0.10, 0.25, 0.50\}$ does not include $r < 0.10$; performance at very-low-dose ($r \lesssim 0.05$) is not characterized. (3) The UQ ensemble disagreement can be non-informative when all members fail in correlated ways (e.g., on out-of-distribution anatomical regions); the reliability diagram should be inspected before using the variance for case triage. (4) The method is single-slice 2D; 3D-aware reconstruction may achieve better axial consistency but at substantially higher compute cost. (5) The TV warm start is non-learned; replacing it with a learned warm-start network is a candidate future improvement we deliberately defer to follow-on work.

7 Conclusion and reproducibility

We have presented an open-source reference reconstruction method for low-dose CT, designed not to maximize PSNR but to be *credible, improvable, and permissively licensed*: cross-vendor evaluated, deep-ensemble UQ-calibrated, signal-equivalence-credentialed per ([to be confirmed](#), 2027b), and Apache-2.0 licensed. The method is the seed entry of the permanent PWM Low-Dose CT Challenge leaderboard ([to be confirmed](#), 2028); an L4 cert anchoring the RunBundle CID is registered on the PWM blockchain mainnet. We invite the community to compete by improving any of the four independent axes (denoiser, forward model, iteration schedule, UQ method) without re-deriving the full stack.

Reproducibility commitment. All code (Apache 2.0), pre-trained ensemble weights, pinned Dockerfile, evaluation scripts, and a content-addressed PWM RunBundle are released at https://github.com/integritynoble/low_dose_CT/tree/main/WS-3_reference_method. The exact library versions, random seeds (Supplementary Table S1), and training-compute disclosure permit bit-identical reproduction within IEEE 754 FP tolerance on equivalent hardware. The RunBundle is independently verified by the PWM Low-Dose CT Challenge scoring service before being posted to the leaderboard, providing third-party reproducibility evidence beyond our own claim.

Acknowledgments

[TODO: Fill at submission.]

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